

Q1:

Imagine you are monitoring a system that predicts stock prices. You observe a sudden market crash. How would ADWIN react in this scenario?

Q2:

What type of concept drift (abrupt or gradual) is ADWIN more sensitive to, and why?

Answer 1: ADWIN would detect the abrupt change in the data distribution by identifying a significant difference in the means of its sub-windows. It would then shrink the window by discarding older data to focus on the new distribution.

Answer 2: ADWIN is more sensitive to abrupt drift because it quickly detects significant changes in the distribution within its dynamic window. Gradual drifts may take longer to detect as the window adjusts incrementally.